

Lab 7: Recycle Bin and Anti-forensics

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Image “cfreds_2015_data_leakage_pc.dd” from Lab 5
- Link download webhistory folder:
<https://drive.google.com/file/d/1b0RNxK1vYoAfv8t0ablopIQ-0cF5DbqD/view>

Step 1.

- Examine ‘Recycle Bin’ data in PC.

```
(hainguyen@hainguyen)~[/lab]
$ fls -rd -w 206848 cfreds_2015_data_leakage_pc.dd | grep -Ei "Recycle" | head
r/- * 0: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/desktop.ini
-/r * 74311-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I55Z163
-/r * 74312-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$IXWGVWC
-/r * 74313-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I40295N
-/r * 74395-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$I9W7UMY
-/r * 74398-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RIQGWTT.ini
-/r * 74418-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RJENT64.exe
-/r * 74418-128-4: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RJENT64.exe:Zone.Identifier
-/r * 74620-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RWXD1U3.jpg
-/r * 74625-128-1: $Recycle.Bin/S-1-5-21-2425377081-3129163575-2985601102-1000/$RI3FM2A.jpg
(hainguyen@hainguyen)~[/lab]
$
```

Step 2.

- Use *testdisk* to Recover *\$Recycle.bin*

Recover the image

```
(hainguyen@hainguyen)~[/lab]
$ testdisk cfreds_2015_data_leakage_pc.dd
```

```
hainguyen@hainguyen: ~/lab × + ▾
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB

>[Proceed ] [ Sudo ] [ Quit ]

Note: Some disks won't appear unless you are root user.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and B
detection, and install the latest OS patches and disk drivers.
```

```
hainguyen@hainguyen: ~/lab × + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

```
hainguyen@hainguyen: ~/lab × + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB
CHS 2611 255 63 - sector size=512

[Analyse ] Analyse current partition structure and search for lost partiti
>[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options ] Modify options
[MBR Code] Write TestDisk MBR code to first sector
[Delete ] Delete all data in the partition table
[Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatch
ed.
```

```
hainguyen@hainguyen: ~/lab × + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk cfreds_2015_data_leakage_pc.dd - 21 GB / 20 GiB - CHS 2611 255 63

Partition              Start          End      Size in sectors
1 * HPFS - NTFS        0  32 33      12 223 19      204800 [System Rese
> 2 P HPFS - NTFS      12 223 20     2610 180 2      41734144

[ Type ] [ Boot ] [ List ] >[Undelete] [Image Creation] [ Quit
]                               File undelete
```

– Use testDisk to recover files

```

hainguyen@hainguyen: ~/Lab
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
2 P HPFS - NTFS 12 223 20 2610 180 2 41734144
Deleted files
Previous
522e76.rbf 21-Nov-2014 12:20 405880
522e77.rbf 21-Nov-2014 12:20 405880
522e78.rbf 21-Nov-2014 12:20 405880
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/ARP.ico 23-Mar-2015 16:01 41851
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/CalendarIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/ContactsIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/FindMyiPhoneIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/MailIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/NotesIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/RemindersIcon.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/ShellStreamsShortcut.exe 23-Mar-2015 16:01 287934
Installer/{389768A4-A2B8-4930-ASA2-8169678C9B4C}/iCloudIcon.exe 23-Mar-2015 16:01 41851
MSMAPI/1033/maapisvc.inf 25-Mar-2015 11:19 0
PyWinTypes27.dll 25-Mar-2015 11:21 110880
S-1-5-21-2425377081-3129163575-2985601102-1000/$I40295N 24-Mar-2015 15:51 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I508CBB.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I552163 24-Mar-2015 15:51 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I8VP3XK.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$I9W7UMY 24-Mar-2015 15:51 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$ID013HE.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IFVCH5V.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$II3FM2A.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IIQWTT.ini 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IJEMT64.exe 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IX3BVM.jpg 24-Mar-2015 16:11 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$IXW6VC 24-Mar-2015 15:51 544
S-1-5-21-2425377081-3129163575-2985601102-1000/$RIQWTT.ini 24-Mar-2015 15:57 174
S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe 0
S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe:Zone.Identifier 26
SetupResources.dll 18-Mar-2010 16:16 19288
Next
Use : to select the current file, a to select/deselect all files,
C to copy the selected files, c to copy the current file, q to quit

```

– Choose the destination and copy

```

hainguyen@hainguyen: ~/Lab
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where
S-1-5-21-2425377081-3129163575-2985601102-1000/$RJEMT64.exe will be copied.
Keys: Arrow keys to select another directory
C when the destination is correct
Q to quit
Directory /home/hainguyen/Lab
drwxrwxr-x 1000 1000 4096 17-Jun-2025 10:38
drwx----- 1000 1000 4096 18-Jun-2025 10:59 ..
-rw-rw-r-- 1000 1000 2147483648 17-Jun-2025 10:24 cfreds_2015_data_leakage_pc.7z.001
-rw-rw-r-- 1000 1000 2147483648 17-Jun-2025 10:31 cfreds_2015_data_leakage_pc.7z.002
-rw-rw-r-- 1000 1000 1132827932 17-Jun-2025 10:37 cfreds_2015_data_leakage_pc.7z.003
-rw-rw-r-- 1000 1000 21474836480 21-Apr-2015 14:17 cfreds_2015_data_leakage_pc.dd

```

Quit all

ls S-1-5-21-2425377081-3129163575-2985601102-1000/ -l

```

(hainguyen@hainguyen)~|~/Lab|
$ ls S-1-5-21-2425377081-3129163575-2985601102-1000/ -l
total 64
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$I40295N'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$I508CBB.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$I552163'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$I8VP3XK.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$I9W7UMY'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$ID013HE.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$IFVCH5V.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$II3FM2A.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$IIQWTT.ini'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$IJEMT64.exe'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$IX3BVM.jpg'
-rw-rw-r-- 1 hainguyen hainguyen 544 Mar 24 2015 '$IXW6VC'
-rw-rw-r-- 1 hainguyen hainguyen 174 Mar 24 2015 '$RIQWTT.ini'
-rw-rw-r-- 1 hainguyen hainguyen 26 Dec 31 1969 '$RJEMT64.exe:Zone.Identifier'
(hainguyen@hainguyen)~|~/Lab|
$ |

```

- Match random file names with readable file names in the file

```

(hainguyen@hainguyen) ~/Lab
$ strings -h
Usage: strings [option(s)] [file(s)]
Display printable strings in [file(s)] (stdin by default)

The options are:
-a --all                Scan the entire file, not just the data section [default]
-d --data              Only scan the data sections in the file
-f --print-file-name    Print the name of the file before each string
-n <number>            Locate & print any sequence of at least <number>
                       displayable characters. (The default is 4).
-t --radix={o,d,x}     Print the location of the string in base 8, 10 or 16
-w --include-all-whitespace Include all whitespace as valid string characters
-o                     An alias for --radix=o
-T --target=<BFDFNAME> Specify the binary file format
-e --encoding={S,s,b,l,B,L} Select character size and endianness:
                       s = 7-bit, S = 8-bit, {b,l} = 16-bit, {B,L} = 32-bit
-u --unicode={default|locale|invalid|hex|escape|highlight}
-U {d|l|i|x|e|h}       Specify how to treat UTF-8 encoded unicode characters
-s --output-separator=<string> String used to separate strings in output.
@<file>                Read options from <file>
-h --help              Display this information
-v --version           Print the program's version number

strings: supported targets: elf64-x86-64 elf32-i386 elf32-iamcu elf32-x86-64 pe-i386 pe-x86-64 pei-x86-64 elf64-little elf64-big elf32-little elf32-big pe-
bigobj-x86-64 pe-i386 pdb src64 symbolsrc64 verilog tekhex binary ihex plugin
Report bugs to <https://sourceware.org/bugzilla/>

(hainguyen@hainguyen) ~/Lab
$ strings -e l -f S-1-5-21-2425377081-3129163575-2985601102-1000/\$I508CBB.jpg
S-1-5-21-2425377081-3129163575-2985601102-1000/\$I508CBB.jpg: C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\Hydrangeas.jpg

(hainguyen@hainguyen) ~/Lab
$

```

Examine ‘Recycle Bin’ data in PC :

\$I Name	Timestamp Deleted	Original File (or Directory) Path
\$I508CBB	2015-03-24 15:51:47	C:\Users\informant\AppData\Local\Microsoft\Windows\Burn\Burn\hydrangeas.jpg

1. What actions were performed for anti-forensics on PC at the last day '2015-03-25'?

Step 1.

- Search IE anti-forensics using keyword

[illegible]

Three tasks:

1. Show all IE history
2. Grep date and with key word “anti”

3. Highlight anti in red

```
(hainguyen@hainguyen)~/Lab
$ cat IE11.export/Container* | grep -oEi "Mar 25, 2015.*search\?q=.*anti" | grep anti --color
Mar 25, 2015 14:46:44.771095500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar 25, 2015 14:46:44.752905500 Mar 25, 2015 14:46:44.771005
500 0 0 0 Visited: informant@http://www.bing.com/search?q=anti-forensic+tools&q&s=n&form=QBLH&ppq=anti
Mar 25, 2015 14:46:44.752905500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar 25, 2015 14:46:44.752905500 Mar 25, 2015 14:46:44.752905
500 0 1 0 Visited: informant@http://www.bing.com/search?q=anti-forensic+tools&q&s=n&form=QBLH&ppq=anti
Mar 25, 2015 14:46:44.752905500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:46:44.753428500 Mar 25, 2015 10:46:44.752000000 Mar 25, 2015 14:46:44.752905
500 0 0 86400 :2015032520150326: informant@http://www.bing.com/search?q=anti-forensic+tools&q&s=n&form=QBLH&ppq=anti
(hainguyen@hainguyen)~/Lab
$
```

- Search a list of anti-forensic keywords :

cat webhistory/IE11.export/Container | grep -oEi "Mar 25, 2015.*search\?q=.*(steganography|encryp|cover|log|diab|trace|cover|remove|overwrite|destry|clean)"*

```
(hainguyen@hainguyen)~/Lab
$ cat IE11.export/Container* | grep -oEi "Mar 25, 2015.*search\?q=.*(steganography|encryp|cover|log|diab|trace|remove|overwrite|destry|clean)"
Mar 25, 2015 14:47:51.248205500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:47:51.248728500 Mar 25, 2015 14:47:51.248205500 Mar 25, 2015 14:47:51.248205
500 0 1 0 Visited: informant@http://www.bing.com/search?q=ccleaner&q&s=n&form=QBRE&ppq=cclean
(hainguyen@hainguyen)~/Lab
$
```

Step 2.

- Search IE download history

Search key word “download”, only show first two records for the demonstration purpose :

cat webhistory/IE11.export/Container | grep -oEi "Mar 25, 2015.*download" | head -n2 | grep download --color*

```
(hainguyen@hainguyen)~/Lab
$ cat IE11.export/Container* | grep -oEi "Mar 25, 2015.*download" | head -n2 | grep download --color
Mar 25, 2015 14:48:22.753705500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:41:02.904698900 Mar 25, 2015 14:48:12.400905500 Mar 25, 2015 14:48:22.753705
500 0 0 86400 :2015032520150326: informant@http://www.piriform.com/ccleaner/download
Mar 25, 2015 14:48:39.616205500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:41:13.254998900 Mar 25, 2015 14:48:22.751205500 Mar 25, 2015 14:48:39.616205
500 0 0 86400 :2015032520150326: informant@http://www.piriform.com/ccleaner/download
(hainguyen@hainguyen)~/Lab
$
```

- Search key word “download”, only show last four records for the demonstration purpose

```
(hainguyen@hainguyen)~/Lab
$ cat IE11.export/Container* | grep -oEi "Mar 25, 2015.*download" | tail -n4 | grep download --color
Mar 25, 2015 14:47:29.852305500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:47:29.850328500 Mar 25, 2015 10:47:29.849000000 Mar 25, 2015 14:47:29.852305
500 0 0 86400 :2015032520150326: informant@http://sourceforge.net/projects/eraser/files/Eraser%206.2.0/Eraser%206.2.0.2962.exe/download
Mar 25, 2015 14:48:22.753705500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:41:13.254998900 Mar 25, 2015 10:48:22.751205500 Mar 25, 2015 14:48:22.753705
500 0 0 86400 :2015032520150326: informant@http://www.piriform.com/ccleaner/download
Mar 25, 2015 14:48:12.403405500 Jan 01, 1601 00:00:00.000000000 Apr 20, 2015 14:41:02.904698900 Mar 25, 2015 10:48:12.400905500 Mar 25, 2015 14:48:12.403405
500 0 0 86400 :2015032520150326: informant@http://www.piriform.com/ccleaner/download
Mar 25, 2015 14:47:35.008505500 iedownload M C:\\Users\\informant\\AppData\\Roaming\\Microsoft\\Windows\\IEDownload
(hainguyen@hainguyen)~/Lab
$
```

Step 3.

- Install download programs- *shimcache*

Check *shimcache* to monitor executed programs

```
(hainguyen@hainguyen)~/Lab/RegRipper3.0
$ perl rip.pl -r ~/Lab/SYSTEM -p shimcache | grep 2015-03-25
Launching shimcache v.20220921
LastWrite Time: 2015-03-25 15:31:05Z
C:\\Users\\informant\\Desktop\\Download\\Eraser 6.2.0.2962.exe 2015-03-25 14:47:40 Executed
C:\\Users\\informant\\Desktop\\Download\\Vccsetup504.exe 2015-03-25 14:48:20 Executed
C:\\Users\\INFORM-1\\AppData\\Local\\Temp\\EraserInstall\\Bootstrapper\\dotNetFx40_Full_setup.exe 2015-03-25 14:50:15 Executed
LastWrite Time: 2015-03-25 10:18:30Z
(hainguyen@hainguyen)~/Lab/RegRipper3.0
$
```


- Check if *eraser* or *ccleaner* is managed by Win installer

```
(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ cd ~/Lab/RegRipper3.0
perl rip.pl -r ~/Lab/SOFTWARE -p installer | grep -Ei "eraser|cclearn" -A 1 -B 2

Launching installer v.20200517
Key       : 1F782E6C74E20F54BB15498F51FCBF84
LastWrite: 2015-03-25 14:57:31Z
20150325 - Eraser 6.2.0.2962 6.2.2962 (The Eraser Project)

(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ |
```

- Check Software registry hive

```
rip.pl -r SOFTWARE -p uninstall | grep 2015-03-25 -A 2 -B 2
```

```
rip.pl -r NTUSER_informant.DAT -p uninstall | grep 2015-03-25 -A 2 -B 2
```

```
(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ perl rip.pl -r ~/Lab/SOFTWARE -p uninstall | grep 2015-03-25 -A 2 -B 2
Launching uninstall v.20200525
Microsoft\Windows\CurrentVersion\Uninstall

2015-03-25 14:57:31Z
Eraser 6.2.0.2962 v.6.2.2962

2015-03-25 14:54:33Z
Microsoft .NET Framework 4 Extended v.4.0.30319

2015-03-25 14:54:06Z
Microsoft .NET Framework 4 Extended v.4.0.30319

2015-03-25 14:52:06Z
Microsoft .NET Framework 4 Client Profile v.4.0.30319

2015-03-25 14:51:39Z
Microsoft .NET Framework 4 Client Profile v.4.0.30319

2015-03-25 10:15:21Z
DXM_Runtime
MPlayer2

(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ |
```

- Check *NTUSER_informant.DAT* registry hive

```
(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ perl rip.pl -r ~/Lab/NTUSER_informant.DAT -p uninstall
Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall keys from Software, NTUSER.DAT hives
Uninstall
```

Nothing found

- Check app paths registry

Check Software apppaths

```
hainguyen@hainguyen: ~/lab x + v
(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ perl rip.pl -r ~/Lab/SOFTWARE -p apppaths | grep 2015-03-25 -A 2 -B 2
Launching apppaths v.20200511
(NTUSER.DAT, Software) Gets content of App Paths subkeys

2015-03-25 14:57:31Z
Eraser.exe - C:\Program Files\Eraser\Eraser.exe
2015-03-25 11:14:17Z
cmmgr32.exe -
2015-03-22 15:19:29Z

(hainguyen@hainguyen)-[~/Lab/RegRipper3.0]
$ |
```


Check NTUSER_informant apppaths

```
(hainguyen@hainguyen)-[~/lab/RegRipper3.0]
$ perl rip.pl -r ~/lab/NTUSER_informant.DAT -p apppaths
Launching apppaths v.20200511
apppaths v.20200511
(NTUSER.DAT,Software) Gets content of App Paths subkeys

(hainguyen@hainguyen)-[~/lab/RegRipper3.0]
$ |
```

Step 4

- Execute two programs : *userassist*

```
(hainguyen@hainguyen)-[~/lab/RegRipper3.0]
$ perl rip.pl -r ~/lab/NTUSER_informant.DAT -p userassist | grep "2015-03-25" -A 1 --color
Launching userassist v.20170204
2015-03-25 15:20:47Z
{1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\xpsrchvw.exe (1)
2015-03-25 15:24:48Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Microsoft Office\Office15\WINWORD.EXE (4)
2015-03-25 15:21:30Z
{7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\Google\Drive\googledrivesync.exe (1)
2015-03-25 15:15:50Z
{6D809377-6AF0-444B-8957-A3773F02200E}\CCleaner\CCleaner64.exe (1)
2015-03-25 15:12:28Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Eraser\Eraser.exe (1)
2015-03-25 14:57:56Z
C:\Users\informant\Desktop\Download\ccsetup504.exe (1)
2015-03-25 14:50:14Z
C:\Users\informant\Desktop\Download\Eraser 6.2.0.2962.exe (1)
2015-03-25 14:46:05Z
Microsoft.InternetExplorer.Default (5)
2015-03-25 14:42:47Z
Microsoft.Windows.MediaPlayer32 (1)
2015-03-25 14:41:03Z
{6D809377-6AF0-444B-8957-A3773F02200E}\Microsoft Office\Office15\OUTLOOK.EXE (5)
2015-03-25 15:21:30Z
{0139D44E-6AFE-49F2-8690-3DAFCAE6FFB8}\Google Drive\Google Drive.lnk (1)
2015-03-25 15:15:50Z
C:\Users\Public\Desktop\CCleaner.lnk (1)
2015-03-25 15:12:28Z
C:\Users\Public\Desktop\Eraser.lnk (1)
2015-03-25 14:46:05Z
{9E3995AB-1F9C-4F13-B827-48B24B6C7174}\TaskBar\Internet Explorer.lnk (5)
2015-03-25 14:42:47Z
{9E3995AB-1F9C-4F13-B827-48B24B6C7174}\TaskBar\Windows Media Player.lnk (1)
2015-03-25 14:41:03Z
{0139D44E-6AFE-49F2-8690-3DAFCAE6FFB8}\Microsoft Office 2013\Outlook 2013.lnk (5)
(hainguyen@hainguyen)-[~/lab/RegRipper3.0]
$ |
```

Step 5

Search for deleted .exe files

Only search for 2015-03-25 from Download folder

```
fls -rd -o 206848 cfreds_2015_data_leakage_pc.dd -l | grep -i Users/informant/Desktop/Download/
| grep 2015-03-25 --color
```

```

(hainguyen@hainguyen)~/Lab/RegRipper3.0
$ fls -sd -o 206848 ~/Lab/cfreds_2015_data_leakage_pc.dd -l \
| grep -i 'Users/informant/Desktop/Download/' \
| grep 2015-03-25 --color

-r * 75186-128-4: Users/informant/Desktop/Download/Eraser 6.2.0.2962.exe 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2015-03-25 1
0:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 8317032 0 0
-r * 75186-128-5: Users/informant/Desktop/Download/Eraser 6.2.0.2962.exe:Zone.Identifier 2015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 2
015-03-25 10:47:40 (EDT) 2015-03-25 10:47:40 (EDT) 26 0 0
-r * 75186-128-4: Users/informant/Desktop/Download/ccsetup504.exe 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28
(EDT) 2015-03-25 10:48:28 (EDT) 5344528 0 0
-r * 75186-128-5: Users/informant/Desktop/Download/ccsetup504.exe:Zone.Identifier 2015-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 2015
-03-25 10:48:28 (EDT) 2015-03-25 10:48:28 (EDT) 26 0 0
(hainguyen@hainguyen)~/Lab/RegRipper3.0
$

```

- Only search for deleted .exe from \$UsnJrnl on 2015-03-25

```

(hainguyen@hainguyen)~/Lab/RegRipper3.0/UsnJrnl2Csv
$ grep -Ei "\.exe.*delete" UsnJrnl2Csv/UsnJrnl_2020-11-28_12-15-14.csv
2015-03-22 15:15:32.602087,conhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:32.602087,conhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:32.602087,conhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:32.602087,conhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,instnm.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,setup16.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,user.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,instnm.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,setup16.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.460089,user.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,instnm.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,setup16.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,user.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,instnm.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,setup16.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.475689,user.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:33.647289,DismHost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:43.069706,invalidatefntcache.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:43.069706,invalidatefntcache.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:43.069706,invalidatefntcache.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:43.069706,invalidatefntcache.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:43.132107,DismHost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:50.729319,taskhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:50.729319,taskhost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE
2015-03-22 15:15:51.618521,DismHost.exe,NOT_CONTENT_INDEXED,FILE_DELETE CLOSE

```

What actions were performed for anti-forensics on PC at the last day ‘2015-03-22’?

Timestamp	Behavior	Description
2015-03-22 15:15:32	File Deletion	Nhiều bản ghi cho conhost.exe bị xóa – có thể là che giấu trình vỏ lệnh
2015-03-22 15:15:33	File Deletion	Các file như instnm.exe, setup16.exe, user.exe bị xóa
2015-03-22 15:15:33	File Deletion	Các file bị xóa nhiều lần, có thể dùng nhiều kỹ thuật để chắc chắn xóa sạch
2015-03-22 15:15:33	File Deletion	DismHost.exe bị xóa – đây là công cụ quản lý ảnh hệ điều hành
2015-03-22 15:15:43	File Deletion	invalidatefntcache.exe bị xóa nhiều lần – có thể là công cụ được giả lập
2015-03-22 15:15:50	File Deletion	taskhost.exe bị xóa – là host cho tiến trình Windows task

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 7.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 7 From YOUR NAME**", replacing "YOUR NAME" with your real name.